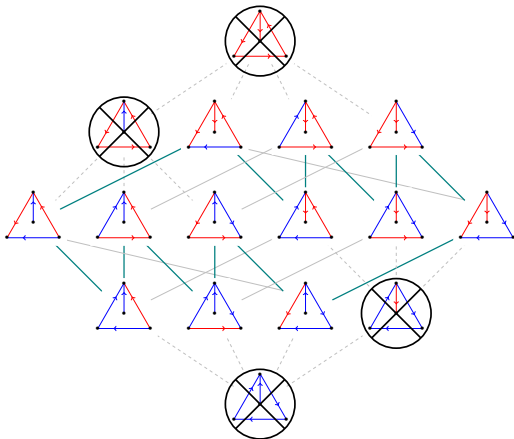


On Hamiltonicity of graphs of acyclic orientations

Germain Poullot with Leonie Mühlherr



2 October 2026, Barcelona

arXiv:2609.02249



1 Graph of acyclic orientations and Gray codes

- First example
- The good old Gray code
- AO-Hamiltonicity

2 Non-AO-Hamiltonian graphs

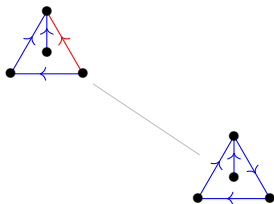
- Divisibility criterion
- Counting acyclic orientations modulo 4
- New results

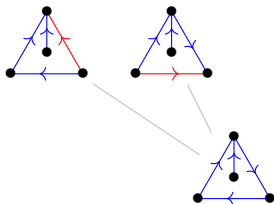
3 AO-Hamiltonian graphs

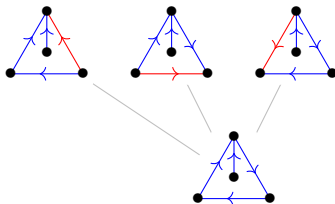
- Towards (open) ear decomposition?
- Replacing patterns for gluing one even path
- New results

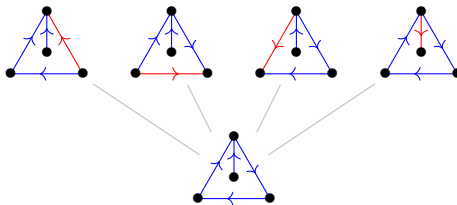
Graph of acyclic orientations and Gray codes

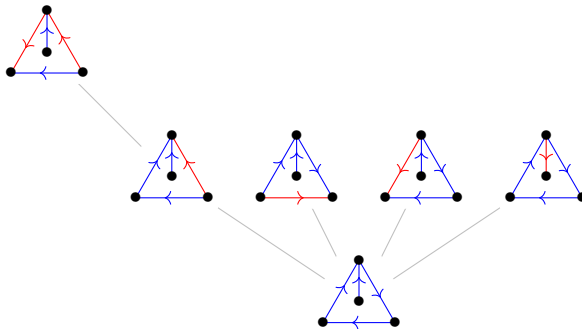


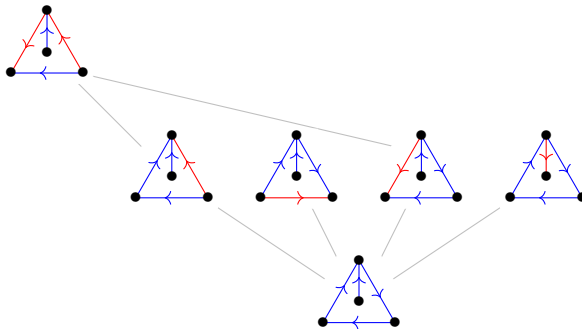


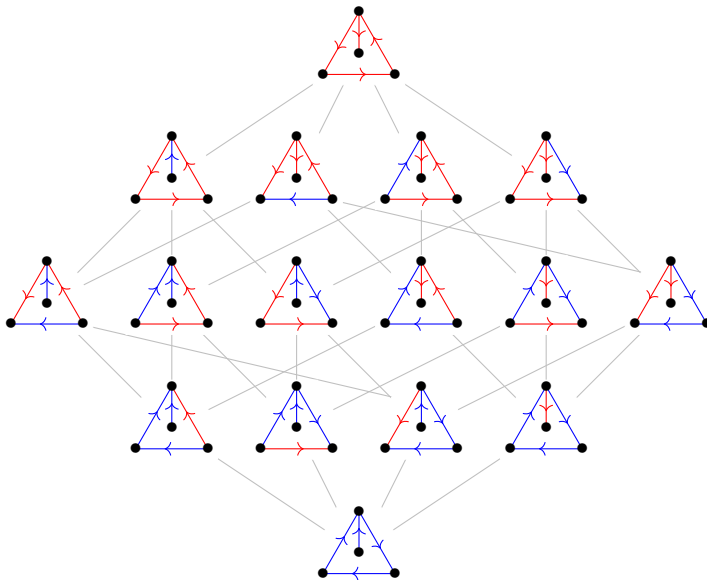


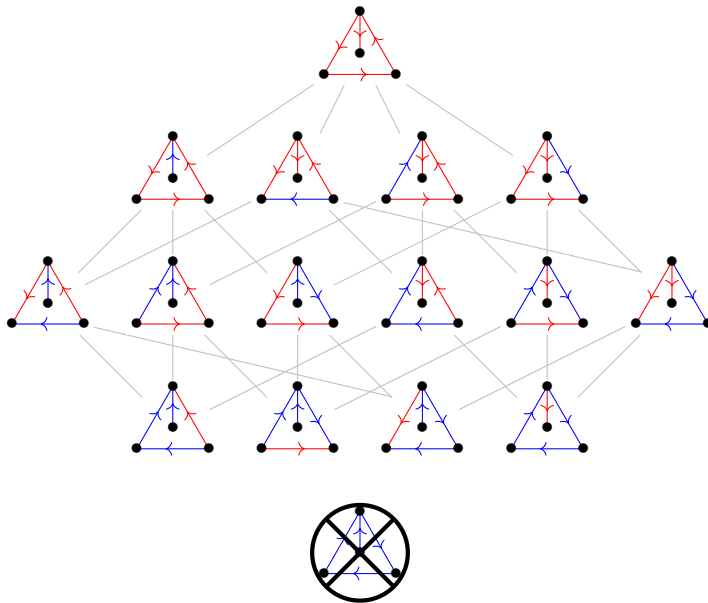


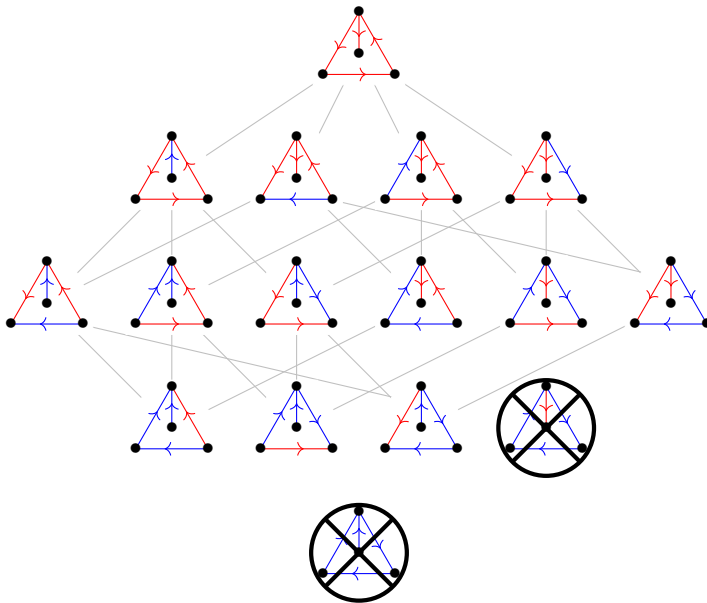


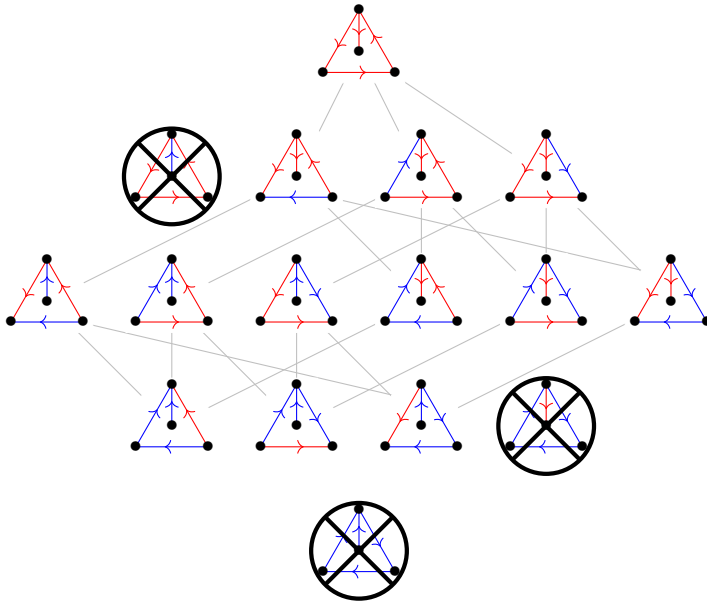


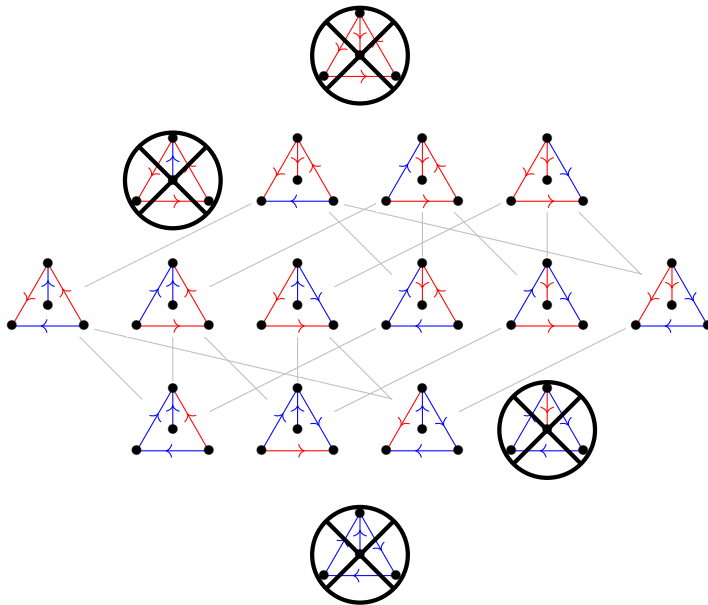


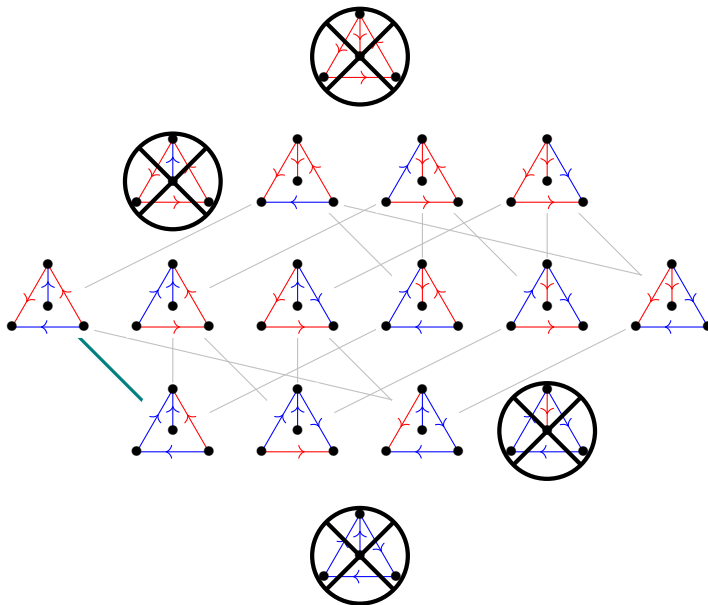


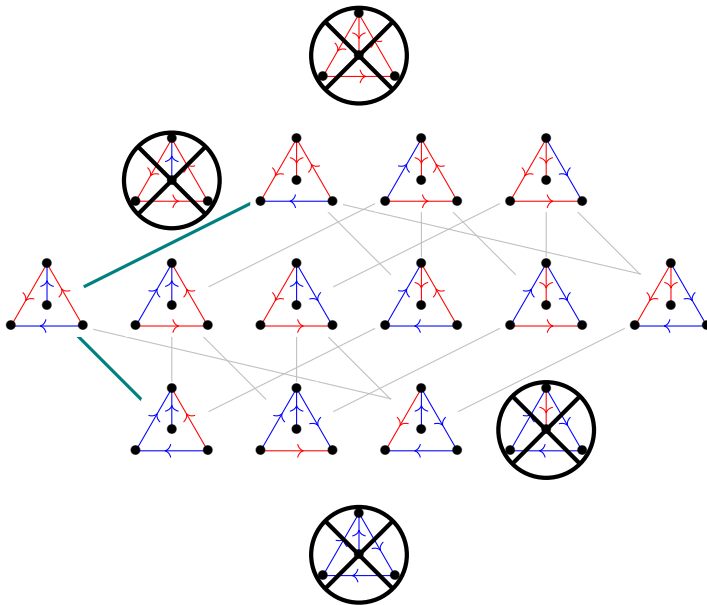


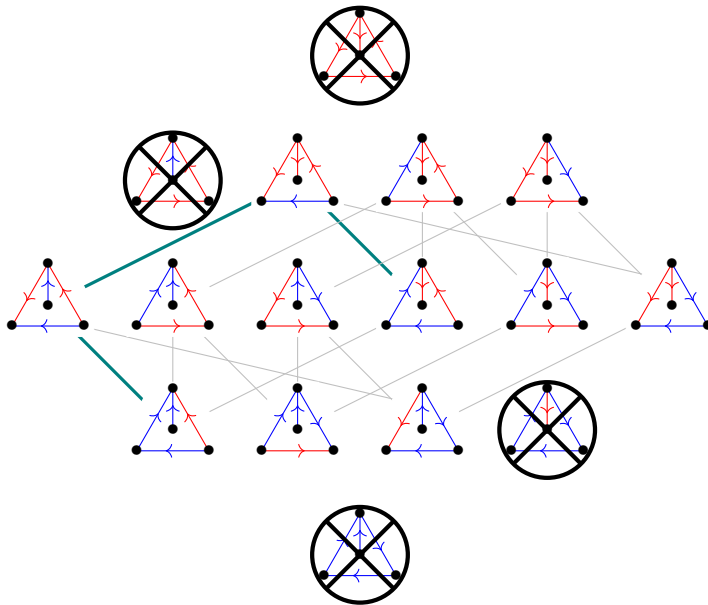


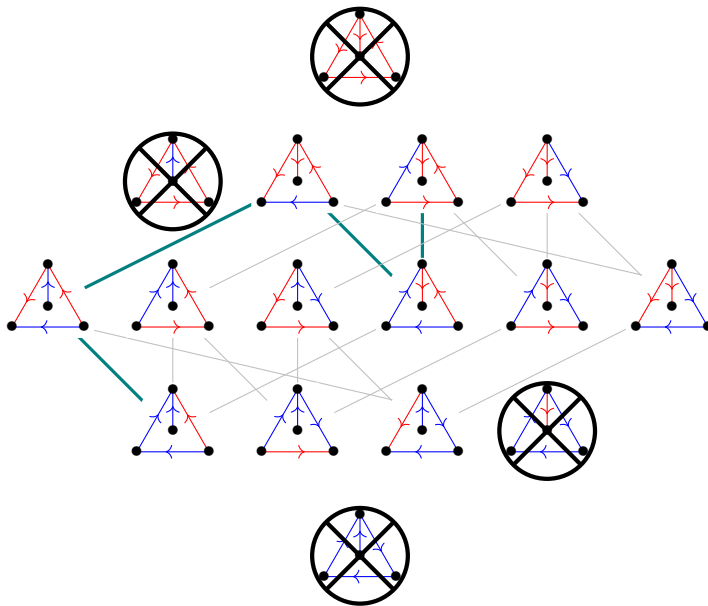


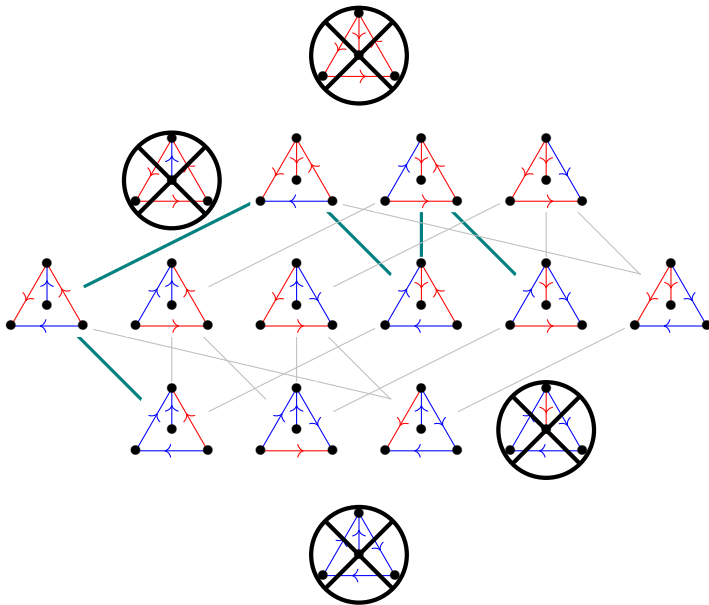


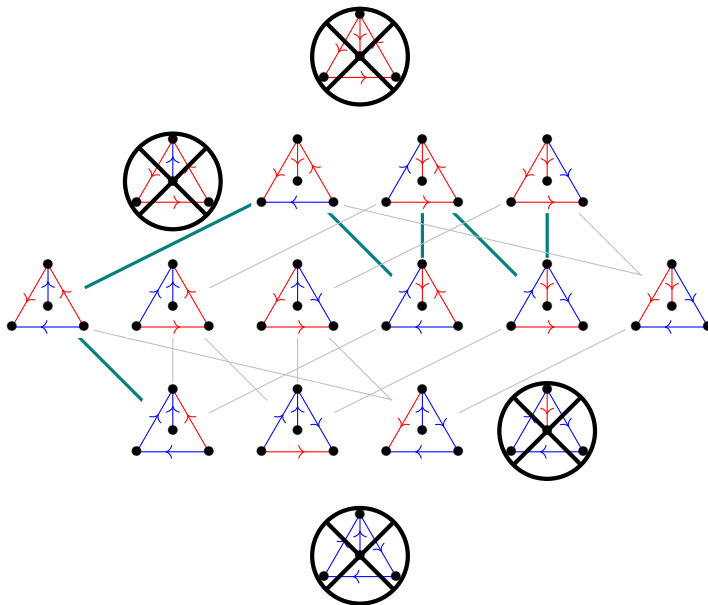


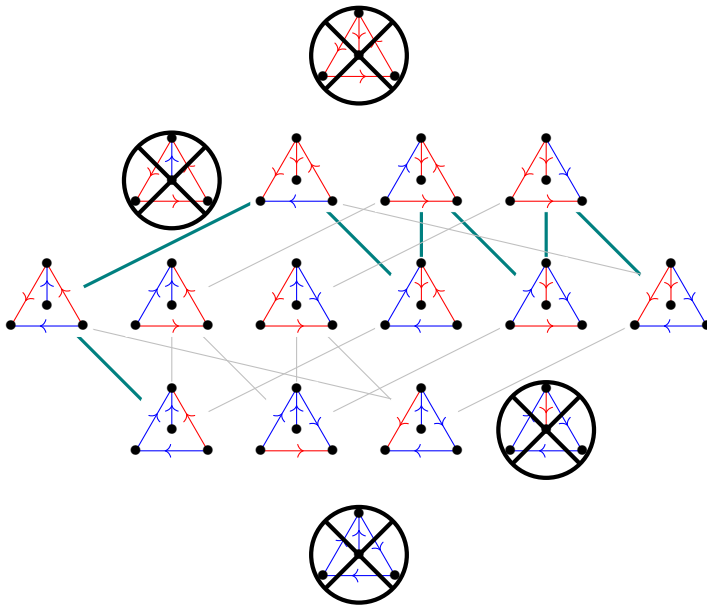


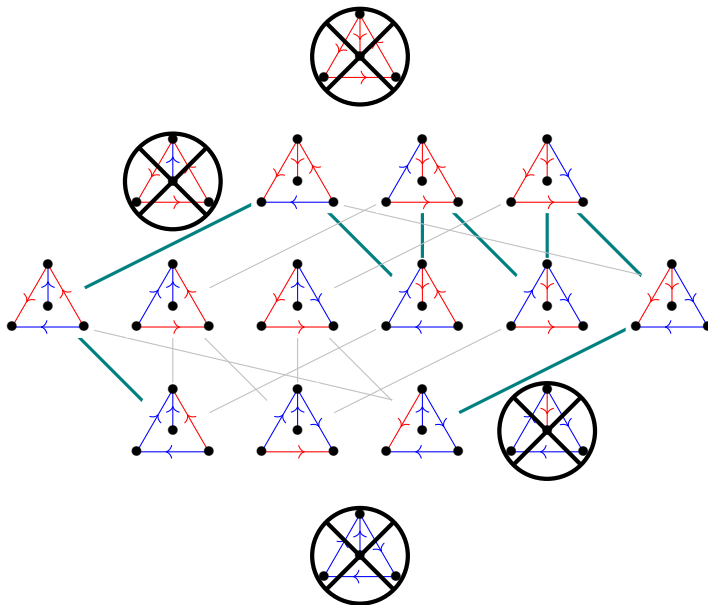


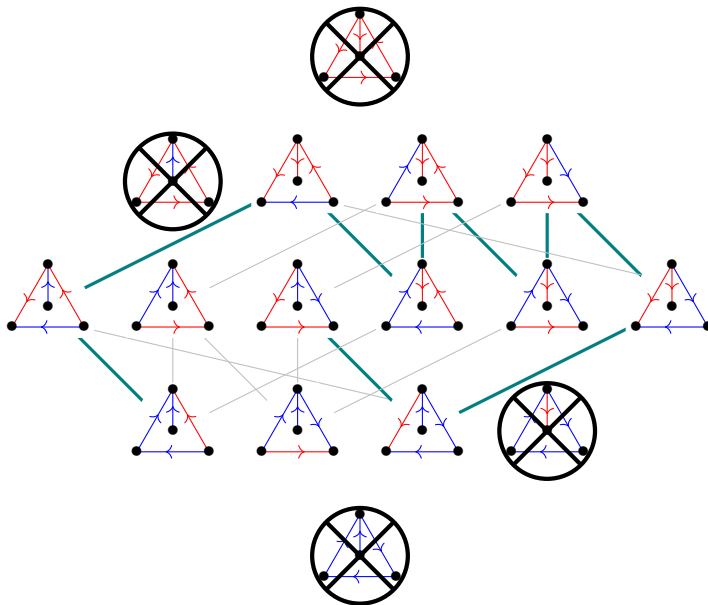


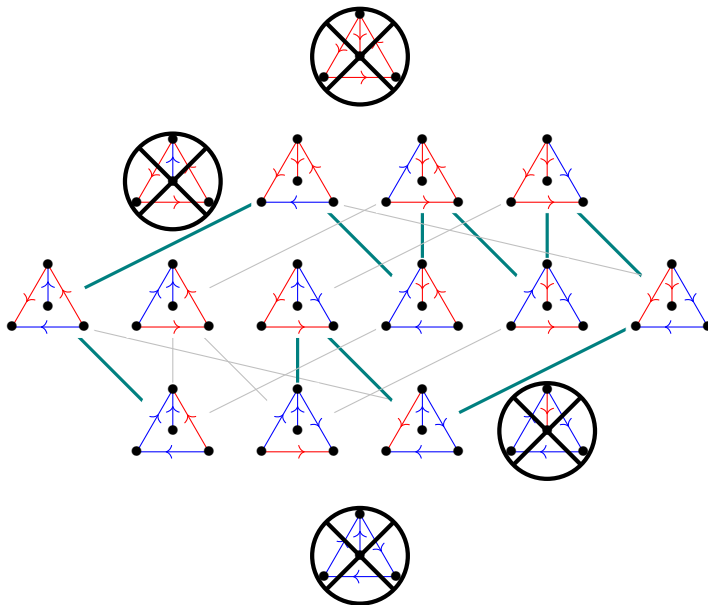


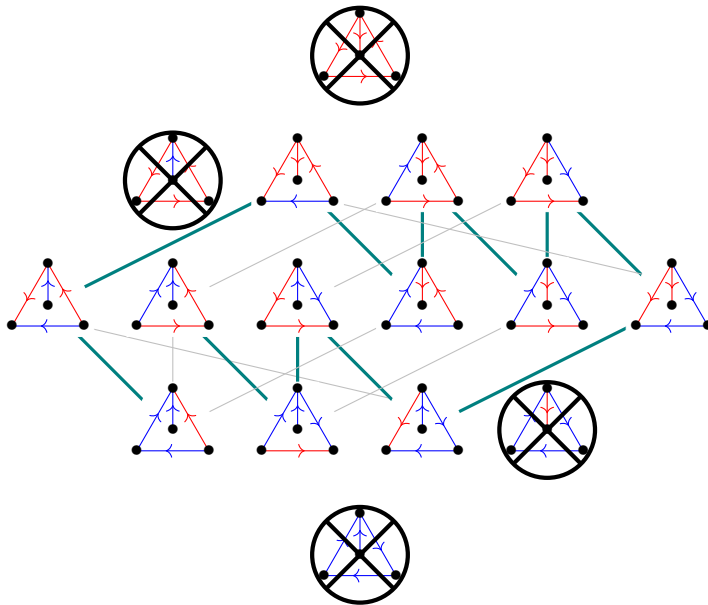


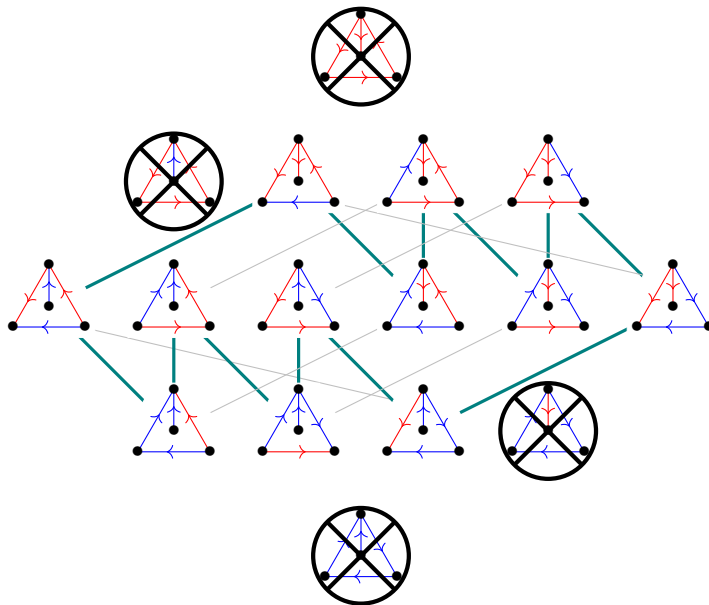


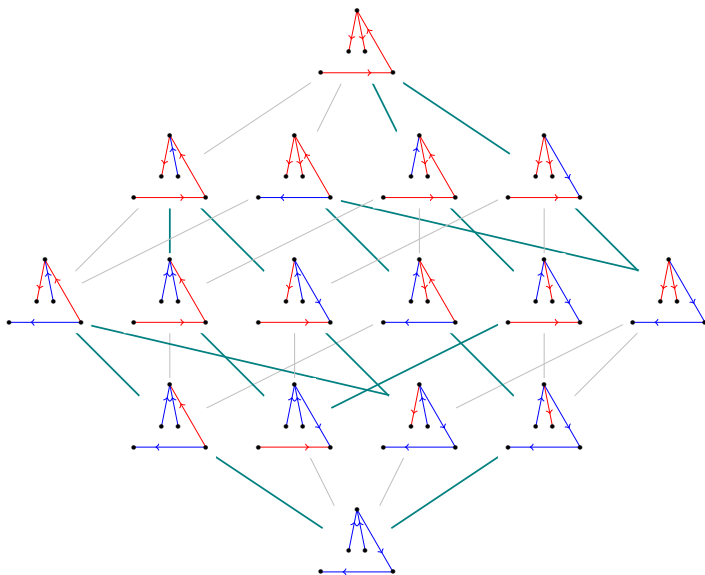


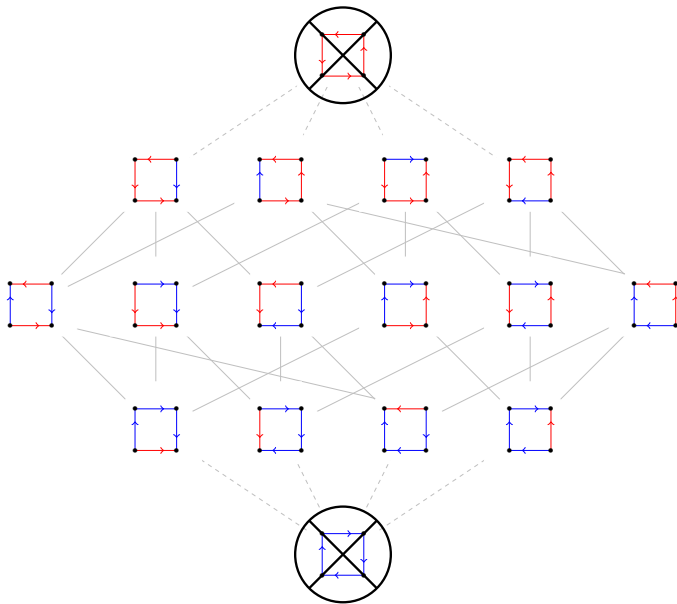




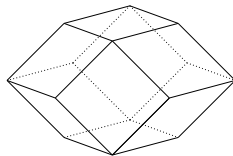
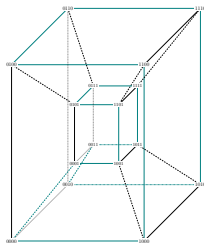
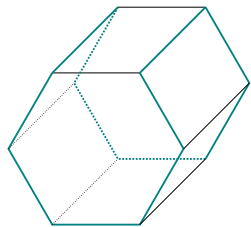
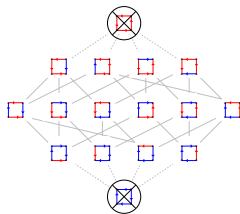
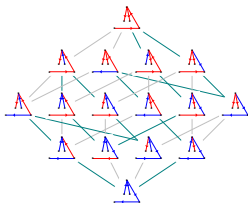
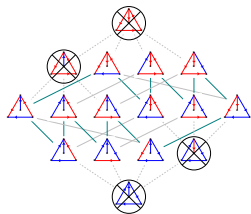








$\mathcal{AO}$ -graph and graphical zonotopes



0000 Switch 1 bit per step

0001

0011

0010

0110

0111

0101

0100

1100

1101

1111

1110

1010

1011

1001

1000

Gray codes

0000	Switch 1 bit per step
0001	No error correction needed
0011	
0010	
0110	
0111	
0101	
0100	
1100	
1101	
1111	
1110	
1010	
1011	
1001	
1000	

Gray codes

0000	Switch 1 bit per step
0001	No error correction needed
0011	Time error resilient
0010	
0110	
0111	
0101	
0100	
1100	
1101	
1111	
1110	
1010	
1011	
1001	
1000	

Gray codes

0000	Switch 1 bit per step
0001	No error correction needed
0011	Time error resilient
0010	Implementable on the hardware. . .
0110	
0111	
0101	
0100	
1100	
1101	
1111	
1110	
1010	
1011	
1001	
1000	

Gray codes

0000

0001

0011

0010

0110

0111

0101

0100

1100

1101

1111

1110

1010

1011

1001

1000

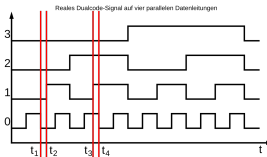
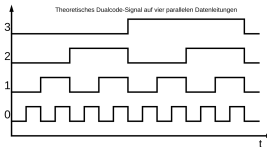
Switch 1 bit per step

No error correction needed

Time error resilient

Implementable on the hardware...

⇒ I gained a factor 1000 in implementations



Wikipedia

Gray codes

0000

0001

0011

0010

0110

0111

0101

0100

1100

1101

1111

1110

1010

1011

1001

1000

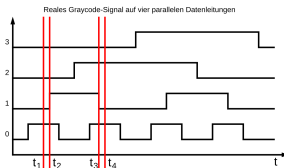
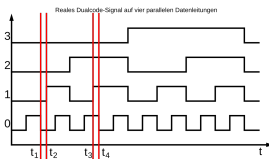
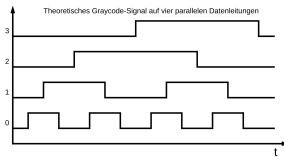
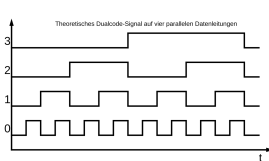
Switch 1 bit per step

No error correction needed

Time error resilient

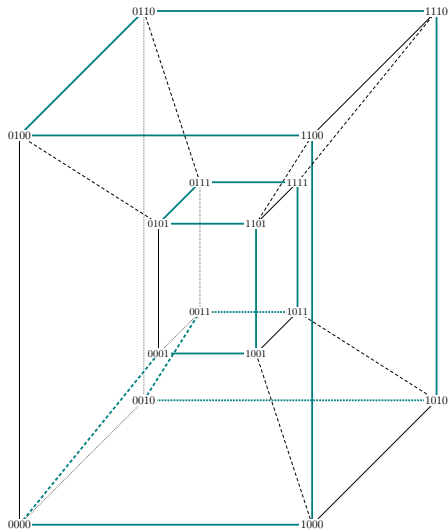
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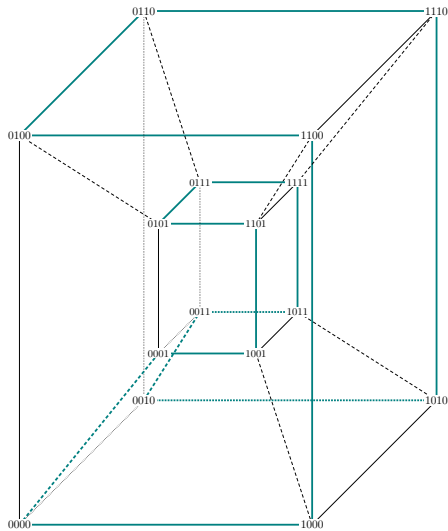


Wikipedia

Gray code = Hamiltonian cycle in cube



Gray code = Hamiltonian cycle in cube



G is $\mathcal{AO}$ -Hamiltonian: there is a Hamiltonian cycle in $\mathcal{AO}(G)$

AO-Hamiltonian:

- Trees

Savage, Squire and West 1993

AO-Hamiltonian:

- Trees
- Cycles odd size

Savage, Squire and West 1993

AO-Hamiltonian:

- Trees
- Cycles odd size

Not AO-Hamiltonian:

- Cycles even size

Savage, Squire and West 1993

AO-Hamiltonian:

- Trees
- Cycles odd size
- Wheel graphs odd size

Not AO-Hamiltonian:

- Cycles even size
- Wheel graphs even size

Savage, Squire and West 1993

AO-Hamiltonian:

- Trees
- Cycles odd size
- Wheel graphs odd size
- Ladder graphs odd size

Not AO-Hamiltonian:

- Cycles even size
- Wheel graphs even size
- Ladder graphs even size

Savage, Squire and West 1993

AO-Hamiltonian:

- Trees
 - Cycles odd size
 - Wheel graphs odd size
 - Ladder graphs odd size
 - Complete graphs
- (Steinhaus–Johnson–Trotter)

Not AO-Hamiltonian:

- Cycles even size
- Wheel graphs even size
- Ladder graphs even size

Savage, Squire and West 1993

AO-Hamiltonian:

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- Chordal graphs

Not AO-Hamiltonian:

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Savage, Squire and West 1993

$\mathcal{AO}$ -Hamiltonian:

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- Cycles odd size
- Wheel graphs odd size
- Ladder graphs odd size
- Complete graphs
(Steinhaus–Johnson–Trotter)
- Chordal graphs

Not $\mathcal{AO}$ -Hamiltonian:

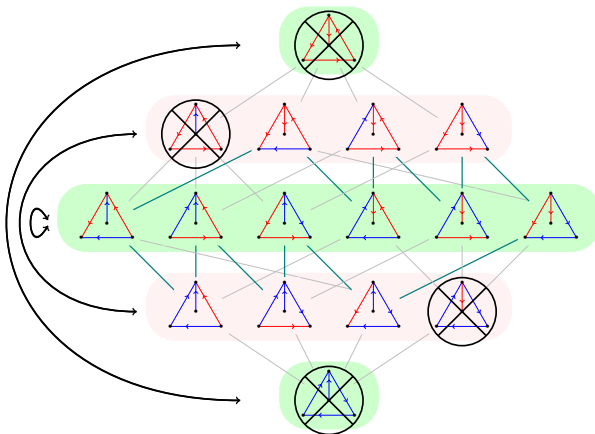
- Cycles even size
- Wheel graphs even size
- Ladder graphs even size
- Complete bipartite
with an even part

Savage, Squire and West 1993

Non- $\mathcal{AO}$ -Hamiltonian graphs

Fundamental involution

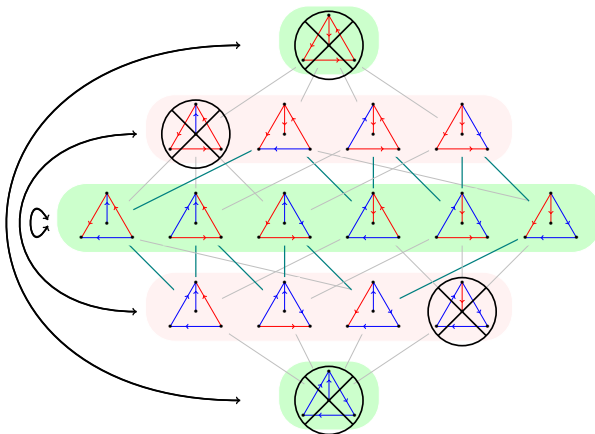
Reverse the orientation of all arcs at once



Involution without fixed point

Fundamental involution

Reverse the orientation of all arcs at once

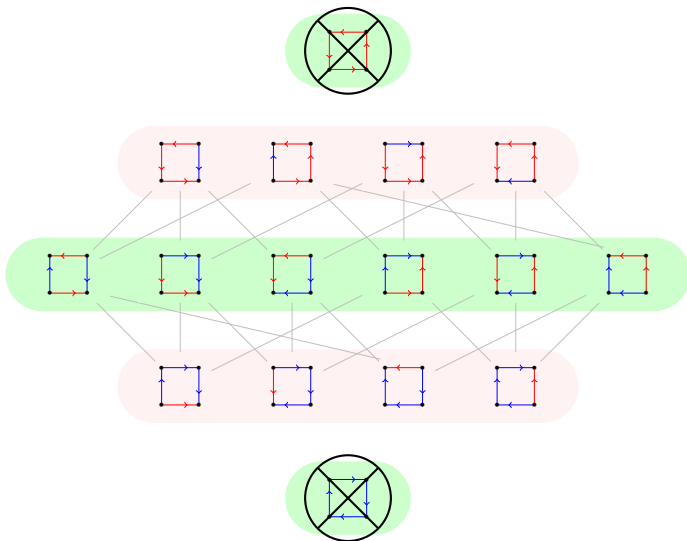


Involution without fixed point

G even nbr edges $\Rightarrow \#\{\text{even AOs}\}$ and $\#\{\text{odd AOs}\}$ are even

Divisibility criterion

8 pink vs. 6 green $\Rightarrow$ **not** AO-Hamiltonian



$\psi(G)$ = number acyclic orientations of G

Lemma (Savage, Squire, West '93)

If G has an even number of edges and is $\mathcal{AO}$ -Hamiltonian, then $\psi(G) \dots$

Proof:

$\mathcal{AO}(G)$ is bipartite.

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Bipartite graph is Hamiltonian $\Rightarrow$ both parts have same size

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$\#\{\text{even AOs}\}$ is even

$\psi(G)$ = number acyclic orientations of G

Lemma (Savage, Squire, West '93)

If G has an even number of edges and is $\mathcal{AO}$ -Hamiltonian, then $\psi(G)$ is divisible by 4.

Proof:

$\mathcal{AO}(G)$ is bipartite.

Bipartite graph is Hamiltonian $\Rightarrow$ both parts have same size

$$\psi(G) = \#\{\text{even AOs}\} + \#\{\text{odd AOs}\} = 2 \cdot \#\{\text{even AOs}\}$$

$\#\{\text{even AOs}\}$ is even

$\psi(G)$ = number acyclic orientations of G

Lemma (Savage, Squire, West '93)

If G has an even number of edges and $\psi(G) \equiv 2 \pmod{4}$, then G is not AO-Hamiltonian.

Proof:

AO(G) is bipartite.

Bipartite graph is Hamiltonian $\Rightarrow$ both parts have same size

$$\psi(G) = \#\{\text{even AOs}\} + \#\{\text{odd AOs}\} = 2 \cdot \#\{\text{even AOs}\}$$

$\#\{\text{even AOs}\}$ is even

Contraction-deletion formula

$G \setminus e$: deletion of the edge e in the graph G

G/e : contraction of the edge e in the graph G

Theorem (Stanley '07 contraction-deletion formula)

e any edge of G :

$$\psi(G) = \psi(G \setminus e) + \psi(G/e)$$

Prisms over trees

$$\psi(\text{prism}(G, \text{path}(u, x, v))) = \psi(\text{prism}(G, \text{path}(u, x))) + \psi(\text{prism}(G, \text{path}(v, x)))$$

Lemma

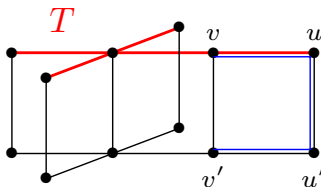
If $\ell \geq 2$ and $uv \in E$, then $\psi(G \oplus \text{path of length } \ell) \equiv \psi(G) \pmod{4}$

Prisms over trees

$$\psi(\text{prism}(G, \text{path}(u,v,x))) = \psi(G \oplus \text{path}(u,v,x)) + \psi(\text{prism}(G, \text{path}(u,v,x)))$$

Lemma

If $\ell \geq 2$ and $uv \in E$, then $\psi(G \oplus \text{path of length } \ell) \equiv \psi(G) \pmod{4}$



Theorem (Mühlherr-P. '26)

$$\psi(T \times K_2) \equiv 2 \pmod{4}$$

T even nbr nodes $\Rightarrow T \times K_2$ **not** $\mathcal{AO}$ -Hamiltonian.

Cartesian product with a cube

Q_n : cube (graph) of dimension n

$$Q_1 = K_2$$

Theorem (Stanley '07 contraction-deletion formula)

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Theorem (Mühlherr-P. '26)

e any edge of G :

$$\psi(G \times Q_n) \equiv \psi((G \setminus e) \times Q_n) + \psi((G/e) \times Q_n) \pmod{4}$$

Careful! $G \mapsto (\psi(G) \pmod{4})$ and $G \mapsto (\psi(G \times Q_n) \pmod{4})$
have same recursion formula, but not same initialization.

Corollary (Mühlherr–P. '26)

*The following graphs are **not** AO-Hamiltonian:*

- Cubes Q_n for $n \geq 2$,

Corollary (Mühlherr–P. '26)

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- *Cubes Q_n for $n \geq 2$,*
- *$T \times Q_n$ for tree T and $n \geq 2$,*

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- Toroidal grid graphs $T_{4,m} := C_4 \times C_m$ for m even,

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- $T \times K_2$ for tree T with even number of nodes,
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- Toroidal grid graphs $T_{4,m} := C_4 \times C_m$ for m even,
- Thick wheel graphs $\text{TW}_n := W_n \times K_2$ for n odd,

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- Thick wheel graphs $\text{TW}_n := W_n \times K_2$ for n odd,
- Any $(T, n_1, \dots, n_s)$ -tree of cycles for any tree T , if n_i even for some i and $\sum_{i=1}^s (n_i - 1)$ is odd...

Corollary (Mühlherr–P. '26)

The following graphs are **not** AO-Hamiltonian:

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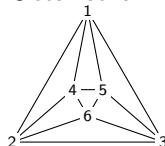
Corollary (Mühlherr–P. '26)

If $\psi(G \times K_2)$ not divisible by 4 and G has an even number of nodes, then $G \times Q_n$ is **not** AO-Hamiltonian for $n \geq 2$.

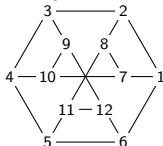
Want to participate?

More graphs, same method, **not** $\mathcal{AO}$ -Hamiltonian:

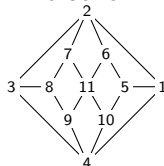
Octahedral



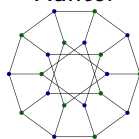
Franklin



Herschel

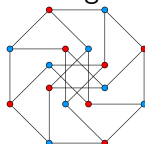


Möbius–Kantor



Wikipedia

Desargues

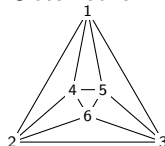


Wikipedia

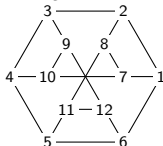
Want to participate?

More graphs, same method, **not** $\mathcal{AO}$ -Hamiltonian:

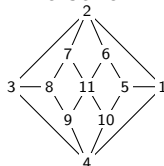
Octahedral



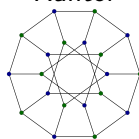
Franklin



Herschel

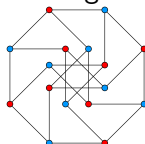


Möbius–Kantor



Wikipedia

Desargues



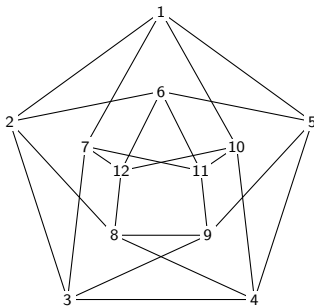
Wikipedia

Open problems: Prove that these are **not** $\mathcal{AO}$ -Hamiltonian:

- The grid graphs $\text{Gr}_{n,m} = P_n \times P_m$ for $m, n \geq 3$ both odd;
- The triangular grid graphs $\text{Tr}_{n,m}$ for $m, n \geq 3$ both odd;
- The generalized Petersen graphs $\text{Pet}(n, k)$ for n even and k odd;
- The complete tripartite graphs $K_{m,n,p}$ for n, m, p not all odd.

Limits of the method

Chvátal graph is **not** $\mathcal{AO}$ -Hamiltonian
but $\psi(G)$ divisible by 4...



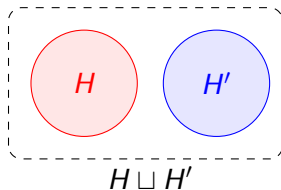
→ Read *acyclic polynomial/signature* in our paper! ^^

AO-Hamiltonian graphs

Theorem

H and H' not connected:

H and H' are $\mathcal{AO}$ -Hamiltonian $\Rightarrow H \sqcup H'$ is $\mathcal{AO}$ -Hamiltonian

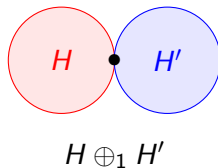
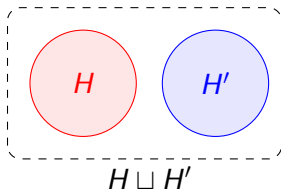


Theorem

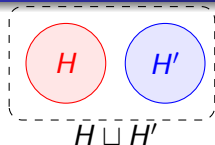
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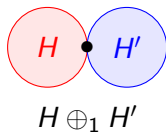
H and H' are $\mathcal{AO}$ -Hamiltonian $\Rightarrow H \oplus_1 H'$ is $\mathcal{AO}$ -Hamiltonian



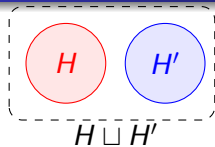
Disconnected graphs and 1-sums



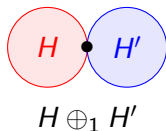
or



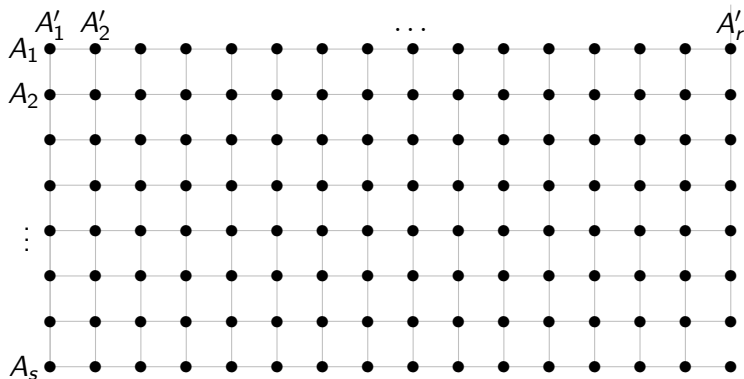
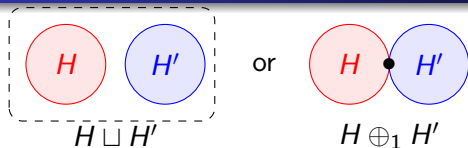
Disconnected graphs and 1-sums



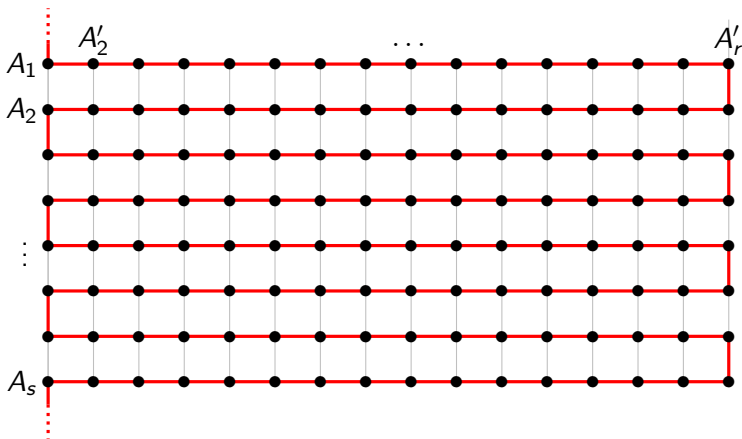
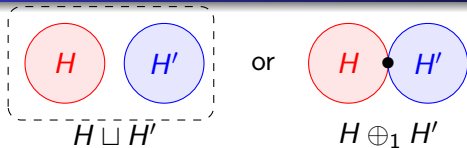
or



Disconnected graphs and 1-sums



Disconnected graphs and 1-sums



(Open) ear decomposition

Theorem ((open) ear decomposition)

Every 2-connected graph admits an (open) ear decomposition, i.e. is obtained by gluing paths iteratively starting on a cycle.

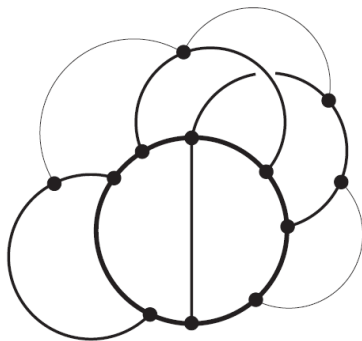
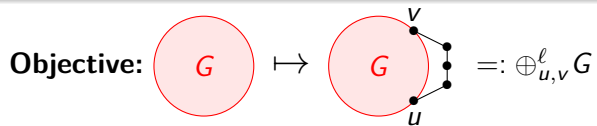
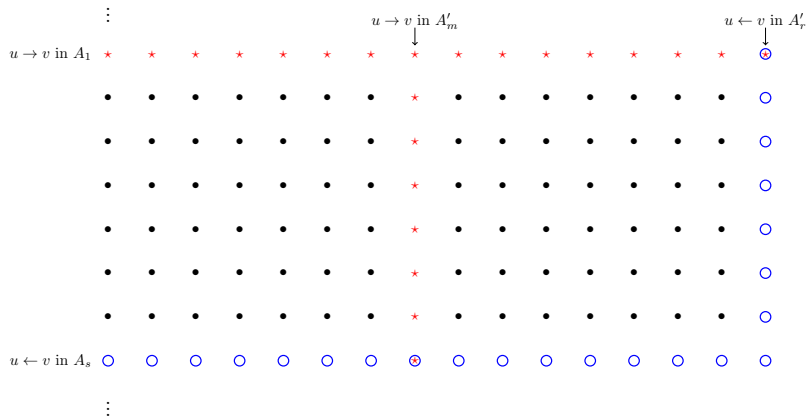
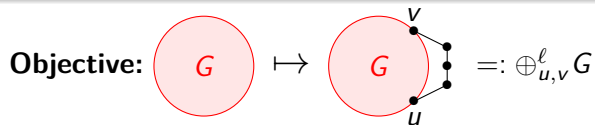


Fig. 3.1.1. The construction of 2-connected graphs
Diestel, *Graph Theory*, Section 3

Replacing patterns

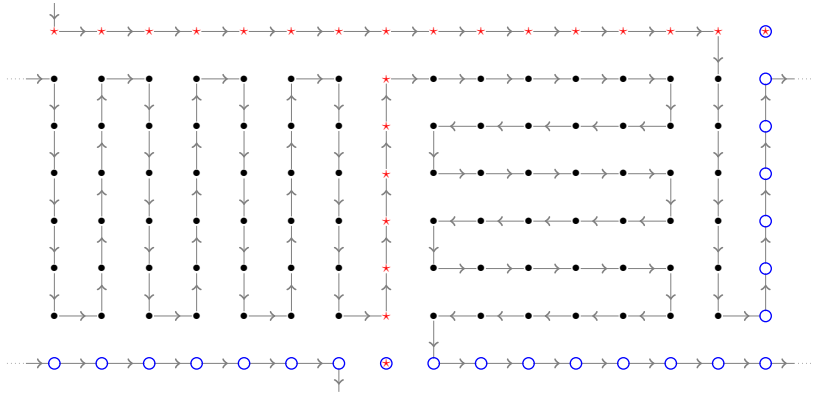


Replacing patterns

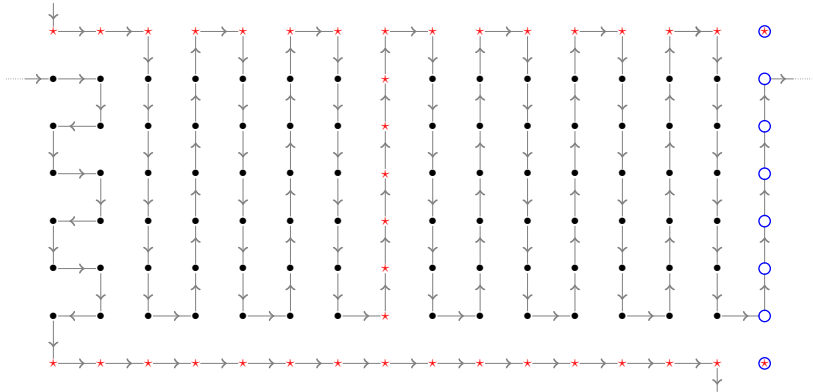


$\star + \circ \Leftrightarrow$ not acyclic

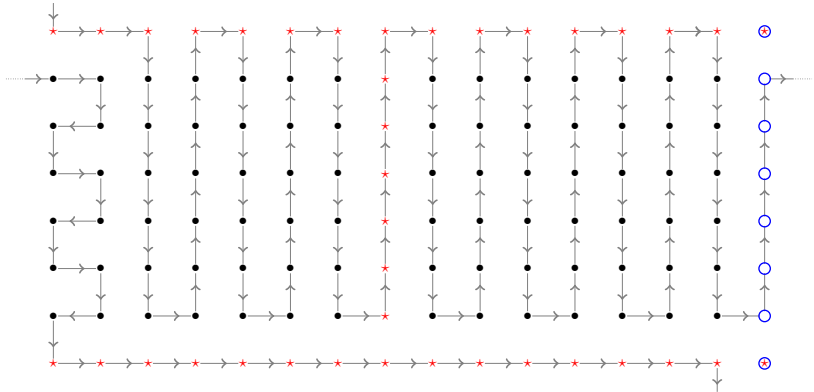
Replacing pattern for gluing one even path



Replacing pattern for gluing one even path



Replacing pattern for gluing one even path



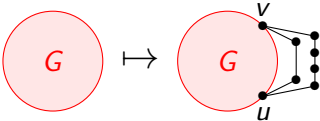
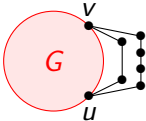
G is *0-even $\mathcal{AO}$ -Hamiltonian* iff G has a $\mathcal{AO}$ -Hamiltonian cycle such that the number of consecutive $\bullet$ is always even

Theorem (Mühlherr–P. '26)

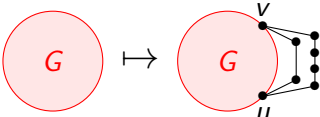
For ℓ even, if G is 0-even $\mathcal{AO}$ -Hamiltonian, then so is $\oplus_{u,v}^{\ell} G$.

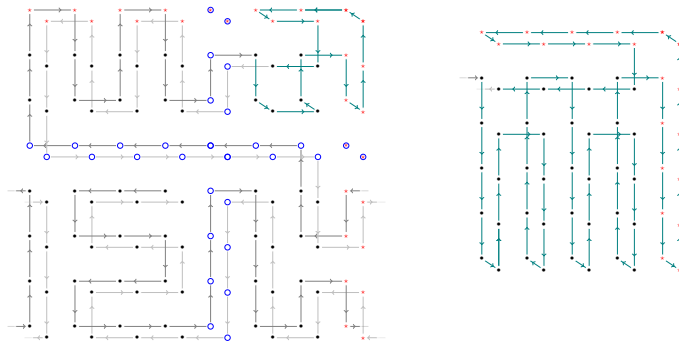
Proof: Find replacing patterns for everything + link them together

Replacing pattern for gluing two odd paths

Objective:  $\mapsto$  $=: \bigoplus_{u,v}^{\ell_1, \ell_2} G$

Replacing pattern for gluing two odd paths

Objective:  $\mapsto \oplus_{u,v}^{\ell_1, \ell_2} G$



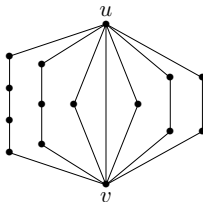
Theorem (Mühlherr-P. '26)

For ℓ_1, ℓ_2 odd, if G is 0-even $\mathcal{AO}$ -Hamiltonian, then so is $\oplus_{u,v}^{\ell_1, \ell_2} G$.

New (positive) results

Corollary (Mühlherr–P. '26)

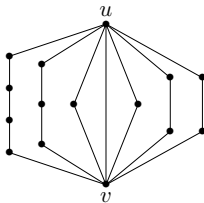
Gluing a multipath with an even number of edges preserves being 0-even $\mathcal{AO}$ -Hamiltonian.



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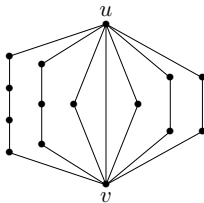
The following are $\mathcal{AO}$ -Hamiltonian:

- *Multipath with odd nbr edges,*

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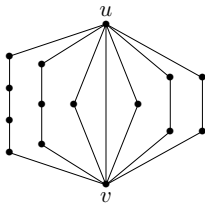
The following are $\mathcal{AO}$ -Hamiltonian:

- *Multipath with odd nbr edges,*
- *Any 2-sum of cycles C_n and C_m if and only if $n + m$ even,*

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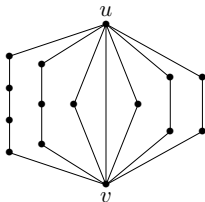
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New (positive) results

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Gluing a multipath with an even number of edges preserves being 0-even $\mathcal{AO}$ -Hamiltonian.



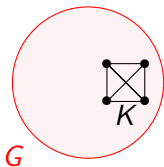
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The following are $\mathcal{AO}$ -Hamiltonian:

- *Multipath with odd nbr edges,*
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- *Any $(T, n_1, \dots, n_s)$ -tree of cycles for any tree T if all n_i are odd,*
- *$G \oplus_{u,v}(\text{even multipath})$ if G is $\mathcal{AO}$ -Hamiltonian and $uv \in G$.*

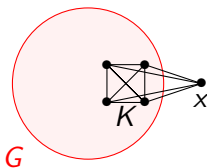
Bonus - Simplicial nodes

Simplicial node x : the neighbors of x form are pairwise adjacent



Bonus - Simplicial nodes

Simplicial node x : the neighbors of x form a pairwise adjacent

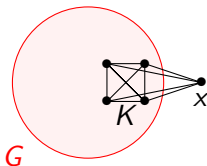


Bonus - Simplicial nodes

Simplicial node x : the neighbors of x form a pairwise adjacent set

Lemma (Savage–Squire–West '93)

Adding a simplicial node preserves $\mathcal{AO}$ -Hamiltonicity.

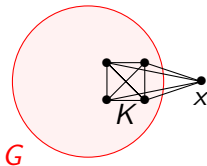


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Theorem (Mühlherr–P. '26)

Adding a simplicial node on a clique K :

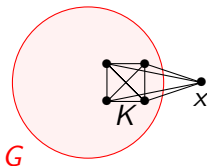
- *preserves 0-even $\mathcal{AO}$ -Hamiltonicity*

Bonus - Simplicial nodes

Simplicial node x : the neighbors of x form a pairwise adjacent

Lemma (Savage–Squire–West '93)

Adding a simplicial node preserves $\mathcal{AO}$ -Hamiltonicity.



Theorem (Mühlherr–P. '26)

Adding a simplicial node on a clique K :

- preserves 0-even $\mathcal{AO}$ -Hamiltonicity
- if $|K|$ is odd, turns $\mathcal{AO}$ -Hamiltonicity into 0-even $\mathcal{AO}$ -Hamiltonicity

Adding simplicial nodes and gluing multipaths

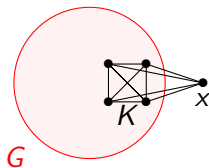
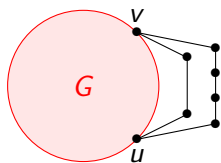
You can play with adding simplicial nodes + gluing multipaths.

Corollary (Mühlherr–P. '26, simplified version)

If G is $\mathcal{AO}$ -Hamiltonian, and if G' is obtained from G by either:

- adding a simplicial node x on an odd clique and gluing an even multipath (not on x); or*

Then G' is $\mathcal{AO}$ -Hamiltonian.



Adding simplicial nodes and gluing multipaths

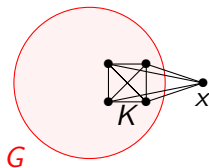
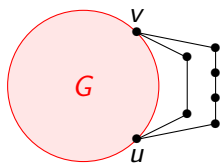
You can play with adding simplicial nodes + gluing multipaths.

Corollary (Mühlherr–P. '26, simplified version)

If G is $\mathcal{AO}$ -Hamiltonian, and if G' is obtained from G by either:

- adding a simplicial node x on an odd clique and gluing an even multipath (not on x); or*
- (optionally) adding a simplicial node and gluing an even multipath on an edge of G ;*

Then G' is $\mathcal{AO}$ -Hamiltonian.



Thank you!

